Ram Analysis of the Plane-Coloring Manuscript M301: Quantifier Preservation, Spectral Transfer, and Exact Local Certificates

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Abstract

We analyze the pinned September 23, 2026 manuscript *The Euclidean plane is not five-colorable*. Its announced unrestricted lower bound is not interchangeable with a measurable or fractional statement. We retain the full quantifiers through its invariant-law, Haar-rigidity, continuous-factor, palette, transition, interface, and angular stages. The private Ram report TR-item-094 supplies a directly applicable finite-witness and exact-certificate framework, not a missing proof of the manuscript's global theorem. We prove explicit local refinements: moment-free stability and certified iteration for the spectral center, a quantitative density collision estimate, an effective circle-averaging tail estimate, and a uniform rational clearance for the final seven-point placement. A complete finite interpolation algebra retains integral coloring constraints, unlike a fractional relaxation. Exact rational and finite exhaustive checks verify the placement, eleven unit edges, and selected angular word assertions. The source's existing Lean declaration is recorded, but its imported proof chain and comparator were not replayed. This is a scoped mathematical analysis with proved local results, not an independent certification or a new proof of the global six-color bound.

Confidential working report. Private source paths and locators are included for local reproducibility; no private corpus text was sent to web search. The catalogue's applicability level is **L5**, not a proof-validity score.

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1 Version, scope, and the central logical distinction

The object of analysis is OpenAI's manuscript at commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`, not an unspecified latest revision. Let U be the graph with vertex set $\mathbb{R}^2$ and edge relation $|x-y|=1$. Its unrestricted target is

$$\forall c:\mathbb{R}^2\longrightarrow\{1,2,3,4,5\}\quad\exists x,y\in\mathbb{R}^2:\quad |x-y|=1,\quad c(x)=c(y). \tag{1}$$

Neither c nor its fibers is assumed measurable. Crossings of edges in a finite drawing are allowed; the planar graph five-color theorem is irrelevant. All compactness arguments below take place in ZFC.

The September 6 private report TR-item-094 records the then-established ordinary interval $5\le\chi(U)\le 7$. The later pinned M301 manuscript *announces* $6\le\chi(U)\le 7$. The former date must not be silently updated into a judgment against the latter, nor may the latter announcement be promoted to independently certified status by this report. Our unconditional conclusions are the elementary local propositions proved here and the explicitly listed finite computations. If the full transfer and geometric obstruction in M301 are correct, its announced lower bound follows. No concrete decisive defect was identified in the reads reported here; that is weaker than global validation, formal replay, or an external consensus claim.

1.1 Reading and formalization scope

The author read all eight mathematical sections, the main file, preamble, and bibliography; an independent local audit read the complete transfer, spectral, and rigidity sections and supplied the center and density refinements below. The author also read the entire actual private TR-item-094, not just its catalogue synopsis. Additional corpus reads are passage-bounded and listed in Section 13. All fourteen build files, including the three figure sources, match the local pinned SHA-256 inventory.

An existing formalization must be acknowledged. The pinned catalogue lists `OAI.Geometry.PlaneColoring.Five`, with target `OAI.EuclideanFiveColor.no_proper_five_coloring`. The inspected solution file imports `BorelTransfer` and `Mesh`, and concludes

$$\neg\exists c:\mathbb{C}\to \operatorname{Fin}5,\ \operatorname{ProperColoring}\ 5\ c.$$

The comparator specification defines properness by exact norm-one pairs, without measurability. Its `sorry` is a challenge placeholder, not evidence of a hole in the submitted solution. The inspected solution has an explicit final proof using `proper_to_borel_weak` and `no_measurable_weak_five_coloring`. Its JSON permits `propext`, `Quot.sound`, and `Classical.choice`; `enable_nanoda` is false. We did not build the Lean project, replay the comparator, inspect every imported theorem, or check the final dependency axioms. Declaration and configuration inspection is not kernel verification. The public catalogue is [2].

2 Four notions of coloring and their quantifiers

Write Col_k for existence of an arbitrary proper k-coloring, and MCol_k for existence with measurable classes. Always $\mathrm{MCol}_k\Rightarrow\mathrm{Col}_k$; the reverse is not a formal consequence of the definitions. For normalized arc measure σ on S^1, the source's weak notion WCol_k asks for a measurable partition $(A_i)_{i=1}^k$ such that

$$\forall R>0,\qquad \sum_{i=1}^{k}\int_{B(0,R)}\int_{S^1}\mathbf{1}_{A_i}(x)\mathbf{1}_{A_i}(x+u)\,d\sigma(u)\,dx=0. \tag{2}$$

Integer $R \geq 1$ suffice by monotonicity. This is product-almost-everywhere exclusion, not exclusion of each pair and not a statement for every direction on every individual circle. Borel representatives of the partition may be chosen. A spatial null modification leaves (2) unchanged: for each fixed u, the exceptional sets at x and $x + u$ have zero area, and Tonelli applies. Such a modification need not preserve pointwise properness.

The source asserts, for every finite k,

$$\mathrm{Col}_k \iff \mathrm{WCol}_k, \qquad \neg\mathrm{WCol}_5.$$

The forward implication is the substantial spectral transfer. The reverse has a short density-plus-compactness proof given below. Even their conjunction does not assert $\mathrm{MCol}_k \Leftrightarrow \mathrm{Col}_k$: a proper coloring obtained by graph compactness need not be measurable.

For a finite graph, fractional coloring instead assigns weights to independent sets. The normalization and vertex-cover constraints are linear; the weights need not describe one integral partition. The probabilistic fields in the source retain *zero products* and permit a measurable positive-label selection. Merely having a small fractional objective would not do this. Restricted polygonal/map-coloring results also retain their own regularity hypotheses. Payne's unit-distance subgraphs emphasize that regularity restrictions can change chromatic behavior [7]; the polygonal result of Sokolov–Voronov is not an unrestricted plane theorem [11].

3 The source dependency chain

3.1 Invariant laws, continuous factors, and the Haar input

Let $F = \overline{\mathbb{Q}} \cap \mathbb{R}$, $E = F(i)$, $K = \{u \in E : |u| = 1\}$, $D = \widehat{E_{\mathrm{disc}}}$, and

$$C = \{j(\xi) : \xi \in \mathbb{R}^2\}, \qquad j(\xi)(z) = e^{i\xi \cdot z}.$$

The group $E \rtimes K$ is countable amenable. A proper labeling of E has a compact orbit closure of proper labelings; finite Følner averages give an invariant probability. None of this gives ordinary spatial continuity. C is Borel by compact exhaustion of frequency space, not assumed closed.

In its Koopman representation the continuous vectors

$$\mathcal{H}_c = \{v : \|T_z v - v\|_2 \to 0 \text{ as } z \in E,\ z \to 0\}$$

are exactly the C-spectral subspace. The two directions use dominated convergence and, conversely, extension of a positive-definite coefficient to $\mathbb{R}^2$, Bochner's theorem, and Fourier uniqueness. Crucially, bounded continuous vectors form a unital conjugation-closed algebra and are stable under Lipschitz truncation. Hence $\mathcal{H}_c = L^2(\mathcal{F}_c)$ for a factor; its projection is conditional expectation, not an arbitrary orthogonal projection. For origin-label indicators,

$$f_i = p_i + b_i, \quad p_i = \mathbb{E}(f_i \mid \mathcal{F}_c), \quad p_i \geq 0, \quad \sum_i p_i = 1.$$

Rotation invariance makes the spectral law of b_i K-invariant and supported off C. The source's Haar-rigidity theorem says that every such normalized law is Haar. Therefore

$$\langle b_i, T_z b_i \rangle = 0 \quad (z \in E \setminus \{0\}).$$

The spectral subspaces are translation invariant, so cross terms vanish. Exact unit exclusion of f_i then yields $\langle p_i, T_u p_i \rangle = 0$ for algebraic unit u. Strong continuity and density of those directions extend this identity to all $u \in S^1$.

Jointly measurable representatives $H_i(x,\omega)$ of $T_x p_i$ exist for almost every x. Nonnegativity, Fubini at both endpoints, and Tonelli over countably many bounded windows give one sample with sum one almost everywhere and all weak collision integrals zero. Select the least index with $H_i > 0$; give arbitrary labels to the remaining spatial null set. This produces (2). The proof does not intersect an uncountable family of probability-one events, nor assume a pointwise action of all real translations.

3.2 What the rigidity theorem actually depends on

The proof is not merely a radial Fourier calculation. A wild invariant law assigns zero mass to each C-coset and to characters continuous on any fixed algebraic line. Divisibility of K rules out its nontrivial finite orbit of positive-mass cosets. An invariant mean on *all* bounded functions on $F/\mathbb{Z}$ defines a positive line functional I; no ordinary measurability of the radial coefficient is assumed. For wild ν,

$$I(\phi(t) f_\nu(at)) = 0 \quad (a \in E \setminus \{0\},\ \phi \in C_c(\mathbb{R})).$$

The smoothing-by-I argument first constructs a strongly continuous vector; the null line event forces it to vanish.

Relative singularity is proved along countable compact-factor towers. A hypothetical positive conditional coset mass is converted to finitely many equivariant centers using the moment-free minimizer in Section 5. Finite phase multisets are encoded by elementary symmetric polynomials. Compact approximants and coefficient bins are fixed *before* the small algebraic step is chosen. All pair-collision bounds hold on a common good fiber before a fiber-dependent subset of indices is selected. Reversing this order would not be justified by the argument.

The compact-tower terminal factor has ergodic relative product. Conditional multiple averages along every Følner sequence then apply to distinct powers of rotations. Surjectivity of $u \mapsto u^n$ for $n \neq 0$ identifies invariant vectors; it does not assert that unweighted images of Følner sets are Følner. The relative difference coefficient

$$m(z) = \|\mathbb{E}(\psi_z \mid Y)\|_2^2 \in [0,1]$$

is nonnegative radial and satisfies

$$m(A)m(B) \le m(|A-B|), \tag{3}$$

$$\mathrm{Avg}_{u \in \Phi_N}\, m(2r|\Re u|) \longrightarrow m(r)^2. \tag{4}$$

The exact identity behind the first inequality is, with $L = \sqrt{AB}$,

$$|A - Bu^2|^2 = (A-B)^2 + 4AB(\Im u)^2 = |-L + (A-B)u + Lu^2|^2.$$

Assuming $m(r) > 0$, syndetic rotations, rational general-position bases of a finite number field, common denominator clearing, and a positive common transversality constant force $m(s) \ge c > 0$ on an interval of positive algebraic radii. The capped-residue lemma uses $w = -\log m$ and bounds a coefficient-length-M integer combination under a cap by $4MH$. The positive interval contradicts the I-line average. Ergodic decomposition and Fourier uniqueness finish the claimed Haar statement. The independent local audit read these arguments completely without identifying a decisive defect; our local center estimates do not prove their structural premises.

3.3 From palettes to connected interfaces

At each fixed center x, weak-* limits of typical circle samples with centers tending to x and radii tending to one define traces $\mathcal{W}_x$. The maximal palette $P_x(e)$ is the essential union of labels having positive trace weight. Countable generating subfamilies suffice for each center. The null set may depend on x; joint measurability in the center is not assumed.

The key sum map is $(e, v) \mapsto x + e + v$. Away from parallel directions its Jacobian is nonzero. On bounded-Jacobian patches, composition of the spatial indicator converges strongly in L^1 while the first angular sample converges weak-*. This proves

$$c(x + e + v) \notin P_x(e) \cup P_x(v) \quad \text{for a.e. } (e, v).$$

One must not replace this with multiplication of two weak-* limits or restriction to an exceptional one-angle fiber. Smooth transport of a first angular endpoint with nonzero derivative yields $P_x(e) \cap P_x(Re) = \varnothing$, where R rotates by $\pi/3$. Binary centers for a fixed label pair are uniformly separated, via six near-identity circle diffeomorphisms, bounded distortion, and propagation of a conull interval. A triple palette would produce infinitely many such centers on a compact circle. Thus $1 \le |P_x(e)| \le 2$ almost everywhere.

An edge ij belongs to the transition graph Γ_x exactly when every ball B centered at x has

$$\limsup_{h \to 0, h \neq 0} D_{ij}(B, h)/|h| > 0.$$

Circle-averaging decay produces a trace avoiding both endpoints of each such edge. Triangle loci reduce to binary centers; four-cycle loci require L^1 continuity of canonical states and uniform support separation; five-cycle loci use support contacts that ban an angular step globally. These are distinct arguments. They establish the claimed closed locally finite cyclic-center locus, not smoothness or finite perimeter of the classes.

Disk averages p_i^ϵ obey $p_i^\epsilon(q) + p_i^\epsilon(z) \le 1$ for every unit pair, by common translations into the typical set. A missing transition edge has $\int p_i^\epsilon p_j^\epsilon = o(\epsilon)$ locally. The Lipschitz bound $O(\epsilon^{-1})$ turns threshold violations into disk covers of vanishing total radii. Thresholding by a fixed $0 < \alpha < 1/5$ and renormalizing puts the remaining probability vector in the one-skeleton of the label simplex. Outside cyclic cores the local graph is a forest, so non-core holes can be filled through trees.

A square-cover obstruction forces a non-null core loop. Reduced graph words, not just cohomology, detect edge midpoints; a connected midpoint preimage crosses an annulus. Hausdorff limits give continua K_j sharing x, each reaching a fixed positive radius, with both edge-endpoint supports absent from

$$\Delta(K_j) = \{z : \min_{q \in K_j} |z - q| < 1 < \max_{q \in K_j} |z - q|\}. \tag{5}$$

The small-hole error is controlled by area at most $4\pi \sum r_B$ of unit-neighborhood annuli. This quantitative order, together with local L^1 convergence, is essential to obtaining support exclusion, not merely exclusion at a sequence of sample points. The extracted cycle need not be a cycle of Γ_x at the limiting center.

3.4 The three angular endings

Selected limiting directions v_j of points of K_j need not be tangents or straight ray segments in K_j. The signs of $e \cdot v_j$ force inner and outer open strips with specified allowed labels. For a five-cycle the signs have runs of lengths two and three; possible palettes are two diagonals whose indices differ by two. A six-position word has steps $\pm 1 \pmod 5$ and antipodal differences $0, \pm 2$. Every triple of steps must be constant, but six identical steps do not close modulo five.

For a four-cycle, effective states are the outsider X and two diagonals U, V. Compact essential supports give a positive cross-word separation. Mixed-parity X words are isolated; dense-set propagation excludes them. All ray signs are split, forcing two antipodal ray pairs. The projective diagonal arcs have total length π, one at most $\pi/2$. A parity change between nearby words would force each arc to contain three positions spaced by $\pi/3$, requiring length greater than $2\pi/3$. Constant parity then contradicts its flip under R.

For a triangle, two outsider labels alternate. Local even unit paths in two open strips give essential local constancy. The essential angular boundary is one six-position orbit, so the outsiders occupy alternating $\pi/3$ sectors. Opposite-sector sums exclude both outsiders on an explicit open polynomial region. A seven-point Moser placement fits there, contradicting three-colorability after one common translation into the typical set. Section 9 verifies this finite terminal certificate exactly.

4 A quantified density-to-collision lemma

Proposition 4.1. *Let $A \subset \mathbb{R}^2$ be measurable, $y = x + u_0$, $|u_0| = 1$, and $0 < h < 2$. Put $\Gamma_h = \{u \in S^1 : |u - u_0| < h\}$ and*

$$a_h = \sigma(\Gamma_h) = \frac{2\arcsin(h/2)}{\pi}, \quad \delta_x = \frac{|B(x,h) \setminus A|}{\pi h^2}, \quad \delta_y = \frac{|B(y,2h) \setminus A|}{4\pi h^2}.$$

Then the local collision integral satisfies

$$\int_{B(x,h)} \int_{\Gamma_h} \mathbf{1}_A(z)\mathbf{1}_A(z+u)\, d\sigma(u)\, dz \geq \pi h^2 a_h [1 - \delta_x - 4\delta_y]_+. \tag{6}$$

If its value is at most $\eta \geq 0$, then $\delta_x + 4\delta_y \geq 1 - \eta/(\pi h^2 a_h)$.

Proof. Choose z uniformly in $B(x,h)$ and independently u uniformly in Γ_h. The first endpoint fails membership with probability δ_x. For each u, $B(x+u,h) \subset B(y,2h)$, so the second failure probability is at most $4\delta_y$. The union bound and nonnegativity give the normalized lower bound. Undo the normalization. The chord formula $|u - u_0| = 2|\sin(\theta/2)|$ gives a_h. The last assertion follows from $[t]_+ \geq t$. □

For example $h = 1$, $\delta_x = \delta_y = 1/20$ give collision mass at least $(\pi/3)(3/4) = \pi/4$. For density-one endpoints, both deficits tend to zero, contradicting zero collision for small h. Therefore density-one sets of each weak color class are unit-independent. They are disjoint and conull by differentiation. Any finite configuration has a common translate inside their union, since only finitely many null sets must be avoided. Thus each finite unit graph is k-colorable, and product compactness proves $\mathrm{WCol}_k \Rightarrow \mathrm{Col}_k$. This complete reverse argument does not use Haar rigidity. A finite empirical collision estimate without rigorous coverage or error control does not establish its premise.

5 Certified spectral centers without moments

For a Borel probability λ on $\mathbb{R}^d$ define

$$J_\lambda(s) = \int \left(\sqrt{1 + |s-t|^2} - \sqrt{1 + |t|^2} \right) d\lambda(t). \tag{7}$$

The integrand is bounded in absolute value by $|s|$. Thus this objective is finite even when $\int |t|\, d\lambda = \infty$.

Theorem 5.1 (Explicit center stability)**.** *Suppose* $\lambda(\overline{B}(0,R)) \geq p > 1/2$*. Set*

$$C_R = R + \sqrt{1+R^2}, \quad B = \frac{pC_R}{2p-1}, \quad \alpha(B') = \frac{p}{[1+(B'+R)^2]^{3/2}} \quad (B' \geq B).$$

The unique minimizer L_λ *satisfies* $|L_\lambda| \leq B$*, and on* $\overline{B}(0,B')$ *one has* $\alpha(B')I \preceq \nabla^2 J_\lambda \preceq I$*. If* $\eta(\overline{B}(0,R)) \geq p$ *too, then*

$$|L_\lambda - L_\eta| \leq \frac{\|\lambda - \eta\|_{\mathrm{TV}}}{\alpha(B)}. \tag{8}$$

Here TV is total variation mass*, twice the usual probability TV distance. For* $|\widehat{s}| \leq B'$ *and a certified gradient value* $\widehat{g}$ *with* $|\widehat{g} - \nabla J_\lambda(\widehat{s})| \leq \epsilon$*,*

$$|\widehat{s} - L_\lambda| \leq \frac{|\widehat{g}| + \epsilon}{\alpha(B')}. \tag{9}$$

If both first moments are finite, $W_1(\lambda,\eta)$ *can replace the numerator in* (8)*.*

Proof. Write $q = \lambda(\overline{B}(0,R))$, the actual mass. The integrand is at least $|s| - C_R$ on the ball and at least $-|s|$ outside it. Consequently $J_\lambda(s) \geq (2q-1)|s| - qC_R$. Coercivity, continuity, and strict convexity give existence and uniqueness. Since $J_\lambda(L_\lambda) \leq 0$,

$$|L_\lambda| \leq \frac{qC_R}{2q-1} \leq \frac{pC_R}{2p-1},$$

where $q/(2q-1)$ is decreasing. Substituting a lower bound for q directly into the global lower bound would not be justified without this step.

Put $v = s - t$. The gradient kernel and Hessian are

$$g_s(t) = \frac{v}{\sqrt{1+|v|^2}}, \qquad H_s(t) = \frac{I}{\sqrt{1+|v|^2}} - \frac{vv^T}{(1+|v|^2)^{3/2}}.$$

Their norms are bounded by one; dominated differentiation needs no moment. For any a, $a^T H_s(t) a \geq |a|^2/(1+|v|^2)^{3/2}$, and $H_s(t) \preceq I$. Integrate the lower bound only over $|t| \leq R$ to get $\alpha(B')$. Strong monotonicity of the gradient along the segment joining the two centers gives

$$\alpha(B)|L_\lambda - L_\eta| \leq |\nabla J_\lambda(L_\eta) - \nabla J_\eta(L_\eta)| \leq \|\lambda - \eta\|_{\mathrm{TV}}.$$

The residual bound follows by the same argument against L_λ and the triangle inequality. The map $t \mapsto g_s(t)$ is 1-Lipschitz because its derivative has operator norm at most one. Integrating its difference under a coupling and taking the infimum gives the W_1 version. □

Corollary 5.2 (A controlled approximation module)**.** *On* $\overline{B}(0,B)$*, let* $s_{n+1} = \mathrm{Proj}_{\overline{B}(0,B)}(s_n - \widehat{g}_n)$*, with* $|\widehat{g}_n - \nabla J_\lambda(s_n)| \leq \epsilon_n$*. Writing* $\alpha = \alpha(B)$ *and* $e_n = |s_n - L_\lambda|$*, one has*

$$e_n \leq (1-\alpha)^n e_0 + \sum_{j=0}^{n-1} (1-\alpha)^{n-1-j} \epsilon_j.$$

For a uniform oracle error ϵ*, the limiting upper error is* ϵ/α*.*

Proof. The integrated Hessian between s_n and L_λ has spectrum in $[\alpha, 1]$. Thus $s \mapsto s - \nabla J_\lambda(s)$ contracts distance to the minimizer by at most $1-\alpha$. Projection onto the convex ball is nonexpansive and fixes the minimizer. Add the oracle error and iterate. □

This is conditional on a certified gradient oracle. Abstract measurable disintegration in the source does not supply an effective oracle or a finite quadrature rule. A partition of $|t| \le R_0$ into cells of diameter d_j and known masses w_j does give an elementary gradient enclosure: node replacement costs at most $\sum_j w_j d_j$, while an omitted tail of mass τ costs at most τ (or 2τ if replaced by an arbitrary node). Certified weight errors add their absolute sum. This needs no first moment. With $p = 3/4$, $R = 0$, the coarse values are $B = 3/2$ and $\alpha = 6/(13\sqrt{13})$; a symmetric law with $3/4$ mass at zero and a symmetric heavy tail has center zero even if its first moment is infinite. These estimates refine a construction inside the proof, not the Haar theorem.

6 Circle averaging as an explicit Ram error channel

The exact circle operator is $\mathcal{S}g(x) = \int_{S^1} g(x+u)\, d\sigma(u)$. For the Fourier convention $\widehat{g}(\xi) = \int g(x) e^{-ix\cdot\xi} dx$, its multiplier is $\widehat{\sigma}(\xi) = (2\pi)^{-1} \int_0^{2\pi} e^{i|\xi|\cos t} dt$. Stationary-arc removal and integration by parts give $|\widehat{\sigma}(\xi)|^2 \le C/(1+|\xi|)$ for an absolute constant. Plancherel normalizations can be absorbed into C in the following bound.

Proposition 6.1 (Quantified small-shift estimate)**.** *Let $g \in L^2(\mathbb{R}^2)$, $0 < r = |h| \le 1$, and $M \ge 1$. Then*

$$\|\mathcal{S}(g - g(\cdot + h))\|_2^2 \le C \left(r^2 M \|g\|_2^2 + r \int_{|\xi|>M} |\widehat{g}(\xi)|^2 \, d\xi \right). \tag{10}$$

In particular this norm is $o(\sqrt{r})$. If $g \in H^s$, $0 < s \le 1/2$, then it is at most $C_s r^{1/2+s} \|g\|_{H^s}$.

Proof. The squared multiplier is bounded by $C \min\{r^2|\xi|^2, 4\}/(1+|\xi|)$. On $|\xi| \le M$ it is at most $Cr^2 M$. Globally it is at most Cr, by splitting at $|\xi| = 1/r$. This proves (10). Divide by r, first let $r \to 0$ with M fixed, and then let $M \to \infty$. For the Sobolev assertion the multiplier divided by $(1+|\xi|^2)^s$ is bounded by $C_s r^{1+2s}$, separately below one, between one and $1/r$, and above $1/r$. Integrate and take square roots. □

If a transition set has area at least $c_B r$, Cauchy–Schwarz turns (10) into an explicit mean-avoidance error of order

$$C c_B^{-1/2} \left(rM\|g\|_2^2 + \int_{|\xi|>M} |\widehat{g}(\xi)|^2 d\xi \right)^{1/2}.$$

For two labels, sum their bounds. A certified tail bound would make the source's $o(1)$ selection quantitative. Arbitrary measurable indicator functions do not come with a Sobolev norm or an effective tail rate; neither is introduced as an extra coloring assumption here.

The Ram local-moment identity is applicable to smooth proxy functions, not to derivatives of arbitrary color indicators. For example averaging over a disk of radius ϵ has second moment matrix $\epsilon^2 I/4$, hence on a C^4 function

$$p_f^\epsilon = f + \frac{\epsilon^2}{8}\Delta f + R_4, \qquad |R_4| \le \frac{\epsilon^4}{72} \sup \|D^4 f\|.$$

Odd terms vanish by symmetry, the disk's fourth absolute moment is $\epsilon^4/3$, and Taylor's factor is $1/4!$. For $f(x) = |x|^2$ the formula is exact: $p_f^\epsilon = |x|^2 + \epsilon^2/2$. For a label indicator this Taylor truncation is generally unavailable; the exact positive averaging operator must be retained. Even a positive kernel's second-order differential truncation need not be positive, as the foundation source explicitly illustrates with $1 + \cos(\omega x)$.

7 Exact finite witnesses and discrete information

Theorem 7.1 (Finite and algebraic witness completeness)**.** *For fixed finite k, $\chi(U) > k$ if and only if there is a finite non-k-colorable graph with an injective unit-distance realization in the plane. Such a witness can be chosen with real algebraic coordinates.*

Proof. If every finite graph constraint were satisfiable, the closed edge conditions in $\{1, \dots, k\}^{\mathbb{R}^2}$ would have the finite intersection property. Compactness would give a proper coloring. The reverse follows by restriction. For a finite witness on n vertices, its realization is an existential formula over $\mathbb{Q}$ consisting of its edge equations and $|p_i - p_j|^2 > 0$ for distinct vertices. Transfer between real closed fields makes it realizable over the real algebraic numbers. Its finitely many coordinates lie in one finite real number field. Extra unit edges cause no difficulty for a lower bound. □

This specializes classical graph compactness [3] and real-closed-field transfer, as already proved in private TR-item-094. It yields a semidecision search, not a bound on vertex count, degree, height, or running time. Searching only one number field, or only repeated unit-circle intersections, is not shown complete.

An exact certificate records a defining field and real embedding, coordinate isolations, exact unit-edge identities, and a refutation of the coloring CNF:

$$\bigvee_{a=1}^{k} X_{v,a}, \qquad \neg X_{v,a} \vee \neg X_{v,b} \ (a < b), \qquad \neg X_{v,a} \vee \neg X_{w,a} \ (vw \in E).$$

An independently checked resolution/LRAT-style proof is preferable to a solver's bare UNSAT flag. Nonedges need not be proved unless a reduction uses their absence. Relabeling colors to fix one vertex's color is satisfiability preserving; identifying a rotational orbit as monochromatic is not. An equilateral triangle is vertex-transitive but requires three colors.

7.1 A complete finite moment reformulation

Theorem 7.2 (Integral coloring in an interpolation algebra)**.** *For a finite graph (V, E) and $k \geq 1$, set*

$$\mathcal{A} = \mathbb{R}[x_v : v \in V] / \left\langle \prod_{c=1}^{k} (x_v - c) : v \in V \right\rangle.$$

Let L_c be the univariate Lagrange polynomial for the node $c \in [k]$, $\delta_a = \prod_{v \in V} L_{a_v}(x_v)$ for $a \in [k]^V$, and $q_{vw} = \sum_c L_c(x_v) L_c(x_w)$. The graph is k-colorable if and only if there is a real linear functional $\ell : \mathcal{A} \to \mathbb{R}$ such that

$$\ell(1) = 1, \qquad \ell(f^2) \geq 0 \ (f \in \mathcal{A}), \qquad \ell(q_e) = 0 \ (e \in E).$$

If it is not colorable, there are explicitly constructible sums of idempotent squares H_e such that $1 = \sum_{e \in E} H_e q_e$ in $\mathcal{A}$.

Proof. Evaluation on $[k]^V$ is an isomorphism $\mathcal{A} \simeq \mathbb{R}^{k^{|V|}}$: reduce separately in every variable to degrees below k, then use product Lagrange interpolation. Thus $\delta_a \delta_b = 0$ for $a \neq b$, $\delta_a^2 = \delta_a$, and $\sum_a \delta_a = 1$. Positivity gives weights $w_a = \ell(\delta_a) = \ell(\delta_a^2) \geq 0$ of total one. The edge indicator equals one precisely when $a_v = a_w$. Its zero expectation eliminates every assignment violating that edge. Some positive-weight assignment remains and is proper. Conversely evaluation at a proper assignment is a functional with the required properties.

If every assignment is improper, choose one violated edge $e(a)$ for it. Set $H_e = \sum_{a:e(a)=e} \delta_a = \sum_{a:e(a)=e} \delta_a^2$. Since $q_e \delta_a = \delta_a$ for an assigned violated edge, summing gives the identity. Everything is rational if the interpolation nodes and choices are rational/discrete. □

For one edge at $k = 2$, only $(1, 2)$ and $(2, 1)$ can have positive weights. For a triangle at $k = 2$ every assignment violates an edge, so choosing its first violated edge produces the displayed refutation. The algebra dimension is $k^{|V|}$; full positivity is complete, while bounded-degree moment relaxations need not be. An invertible Ram basis change transports both the multiplication table and positivity cone; retaining only selected moments changes the problem. No general computational speedup is claimed.

7.2 A finite weighted badness certificate

For nonnegative rational edge weights of positive total W, define

$$\beta_k(H, w) = \frac{1}{W} \min_{a \in [k]^{V(H)}} \sum_{vw \in E(H)} w_{vw} \mathbf{1}_{a_v = a_w}.$$

It is positive exactly when the positive-weight edge subgraph is not k-colorable. For any plane coloring, every isometric image of H has weighted monochromatic sum at least $W\beta_k$. Integrating with a normalized finitely additive invariant mean on isometries gives badness at least β_k, because every unit edge has the same averaged indicator. This is a finite sum, so countable additivity is unnecessary. For a measurable periodic coloring, average a uniform translation in a fundamental cell and a uniform rotation; each edge has the same needle frequency, giving the same lower bound. This proves a weighted refinement of the finite-count argument used in the source's positive-badness corollary. For M301's global five-color conclusion it remains conditional on obtaining a non-five-colorable finite witness. The source's compactness-derived $1/|E(H)|$ is existential, not an explicit numerical universal constant supplied by its seven-point graph.

8 Why fractional bounds cannot supply six colors

Proposition 8.1 (An elementary strict fractional ceiling)**.** *For every finite unit-distance graph G in the plane,*

$$\frac{|V(G)|}{\alpha(G)} \le \chi_f(G) \le \frac{8\sqrt{3}}{\pi} < 5.$$

Proof. Let $\Lambda = \mathbb{Z}(2, 0) + \mathbb{Z}(1, \sqrt{3})$ and take the union of open disks of radius $1/2$ about its points. Distances in one disk are less than one; distances between different disks are greater than one, since distinct centers are at least two apart. Its density in a fundamental cell is $\delta = \pi/(8\sqrt{3})$. A uniform cell translation gives every finite vertex membership probability δ in a random independent set. Assign weight $\Pr(I)/\delta$ to every nonempty independent set obtained; each vertex has covering weight one and total weight at most $1/\delta$. Summing any fractional vertex-cover inequalities gives the lower bound $|V|/\alpha$. Finally $\sqrt{3} < 7/4$ and $\pi > 3$ imply $8\sqrt{3} < 14 < 5\pi$. □

Closed disks would invalidate this proof at their boundaries. The result is weaker than optimized packing bounds but completely suffices to block a fractional lower bound above five. It does not preclude $\chi(G) = 6$: an integral/fractional gap can persist. If an approximate dual weighting satisfies $\sum_{v \in I} y_v \le 1 + \eta$ for *all* independent sets, then $\chi_f(G) \ge \sum_v y_v/(1 + \eta)$; sampling some sets is insufficient.

The primary 2026 Dúcz–Varga result gives a finite unit graph of independence ratio below 1/4, hence a fractional bound above four [9]. This improves a relaxation but integer rounding still gives only five. Dúcz's Moser-lattice note constructs geometric four-colorings on that lattice/ring [10]; it is a restriction on that fractional/geometric strategy, not an ordinary four-coloring of the entire plane. Matolcsi–Ruzsa–Varga–Zsámboki's amenability-based predecessor establishes the bound four [8].

9 The seven-point certificate and a new clearance

Use complex coordinates

$$A = (\sqrt{3}+i)/2, \quad B = (\sqrt{3}-i)/2, \quad T = \sqrt{3}, \quad u = (5+i\sqrt{11})/6, \quad \mathcal{G} = \{0, A, B, T, uA, uB, uT\}.$$

The triangles $(0, A, B)$ and (T, A, B) force $0, T$ to agree in a three-coloring, and their rotated copies force $0, uT$ to agree. But $|T - uT|^2 = 3(1+11)/36 = 1$. There are eleven unit edges in all; this is a four-chromatic graph, not a six-chromatic witness.

The source places its vertices at

$$z_g = \frac{35+12i}{37}\left(g + \frac{-290+149i}{250}\right). \tag{11}$$

The multiplier has norm one. Write $z = \xi + i\upsilon$, $l = |z|^2$, and

$$P = \upsilon^2(3\xi^2 - \upsilon^2)^2, \qquad P' = l^3(1 - l/4)(l-1)^2, \qquad F_*(z) = P' - P.$$

The source's open three-label region is $0 < l < 4$, $F_* > 0$. Indeed, writing $z = re^{i\phi} = e^{i(\phi-\delta)} + e^{i(\phi+\delta)}$ with $\delta = \arccos(r/2)$ gives

$$\sin(3(\phi-\delta))\sin(3(\phi+\delta)) = \frac{P - P'}{l^3} < 0.$$

The two summands are in opposite outsider sectors, and the nonsingular sum map transports the almost-everywhere exclusion to an open neighborhood.

g	ξ_0	υ_0	P_+	P'_-
0	-1.2906	0.1876	0.90	1.35
A	-0.6335	0.9414	0.095	0.110
B	-0.3092	-0.0045	0.000002	0.00065
T	0.3478	0.7493	0.023	0.026
uA	-1.1598	1.1790	9.8	19.0
uB	-0.3666	0.5700	0.0022	0.024
uT	-0.2358	1.5614	12.65	12.9

Table 1: Pinned rational placement certificate, replayed with exact fractions. Coordinate half-width $h = 0.00011$; every row has $P < P_+ < P'_- < P'$.

All decimals are rational. Squaring gives $1.73205 < \sqrt{3} < 1.73206$ and $3.31662 < \sqrt{11} < 3.31663$. Midpoint substitution has per-coordinate error at most 0.000005 before placement, including the $\sqrt{33}$ products. The rotation's absolute row sum is $47/37$; the rational midpoint values differ from the displayed centers by less than 0.00005. Thus their coordinate error is less than h. With

$$x_l = (|\xi_0| - h)^2, \; x_u = (|\xi_0| + h)^2, \quad y_l = (|\upsilon_0| - h)^2, \; y_u = (|\upsilon_0| + h)^2,$$

put $l_l = x_l + y_l$, $l_u = x_u + y_u$, and $H = \max(|3x_l - y_u|, |3x_u - y_l|)$. The exact checker verifies the source's rational intermediate factor table and

$$P \le y_u H^2 < P_+, \qquad P' \ge l_l^3(1 - l_u/4)\min(|l_l - 1|, |l_u - 1|)^2 > P'_-.$$

Every radius interval is inside $(0, 4)$ and avoids one. The strict gap shared by the seven points is at least $m_* = 81/125000 = 0.000648$.

Proposition 9.1 (Uniform local placement clearance)**.** *Every open disk $B(z_g, 10^{-7})$, for $g \in \mathcal{G}$, lies in $\{0 < |z|^2 < 4, F_*(z) > 0\}$. Consequently every common translation of the placement by a vector of norm less than 10^{-7} remains in this region and preserves all eleven exact unit edges.*

Proof. The rational radius enclosures imply $0.3 < |z_g| < 1.7$. On $|z| \le 2$, $P = (\Im z^3)^2$ has gradient norm at most $2|z|^3 \cdot 3|z|^2 \le 192$. For $q(l) = l^3(1 - l/4)(l - 1)^2$ on $[0, 4]$,

$$|q'(l)| \le 3 \cdot 16 \cdot 9 + 16 \cdot 9 + 2 \cdot 64 \cdot 3 = 960.$$

Since $|\nabla l| \le 4$, $|\nabla F_*| \le 3840 + 192 = 4032$. Every segment from z_g to a point in its indicated disk remains in $0 < |z| < 2$. The mean-value bound gives $F_*(z) > m_* - 4032 \cdot 10^{-7} > 0$. Common translation preserves differences. □

This is a conservative new explicit refinement of the source's strict membership certificate. Independent vertex perturbations remain in the region but generally lose the exact edges; only rigid motions preserve the unit graph automatically. Under the source's preceding sector conclusions, each disk uses only the triangle labels almost everywhere. Seven translated null exceptions can be avoided by one common translation in the certified disk, giving the final contradiction. The sector conclusion is indispensable.

10 Ram modules, genuine uses, and forbidden shortcuts

10.1 A concrete representation, not a renamed theorem

A useful local record for M301 is

$$\mathcal{R}_{301} = (\mathcal{Q}, \mathcal{S}, \mathcal{P}, \mathcal{I}, \mathcal{A}, \mathcal{C}, \mathcal{E}),$$

where $\mathcal{Q}$ stores quantifiers and exceptional-set conventions; $\mathcal{S}$ the continuous/wild spectral split and its exact zero correlations; $\mathcal{P}$ the palette/transition label constraints; $\mathcal{I}$ graph words and continuum support exclusions; $\mathcal{A}$ exact algebraic coordinates and polynomial predicates; $\mathcal{C}$ the coloring CNF or full interpolation algebra; and $\mathcal{E}$ certified numerical, tail, and residual budgets. This is an auditable decomposition of concrete objects, not an asserted new isomorphism between spectral and coloring problems. Transform coordinates must preserve positivity, multiplication, labels, and the declared physical norm when any of these is used downstream.

TR248 gives $T^*T = I - U^*LU$ on a named finite physical model, with $L = \sum M_p(I - \Pi_p)M_p \ge 0$. A finite lower bound requires $\|U^*LU\| < 1$. Entrywise certified integration errors give $\|E\| \le \|E\|_F$ and reduce a singular margin from $\sqrt{A}$ to $\sqrt{A} - \|E\|$. They do not frame unrestricted L^2 or recover the wild spectral part. If coefficients use a Gram matrix $G > 0$, the physical coordinates are $G^{1/2}c$, not the raw coefficient vector.

For a finite smooth model $f = \sum c_j\phi_j$,

$$|f(z)| \le \big(v(z)^*G^{-1}v(z)\big)^{1/2}\|c\|_G, \qquad v(z) = (\overline{\phi_j(z)})_j.$$

This follows by Cauchy–Schwarz in Gram coordinates and is the required evaluation estimate when an L^2 coefficient bound is to yield a pointwise geometric budget. On all of L^2, point evaluation is not bounded. For exact vertex position errors $\leq \epsilon$, the squared-distance error is at most $4|p_i - p_j|\epsilon + 4\epsilon^2$, by expanding the square. A small error does not certify distance exactly one.

TR190's purely spectral graph filters are permutation equivariant and admit frame reconstruction when $\sum_j |\kappa_j(\lambda)|^2 \geq A > 0$. Polynomial degree r gives at-most-r-hop support. Neither a small scale nor a vertex ordering by itself establishes exact finite-hop support or a unit-triangular inverse. The application source's retained/discarded identity

$$\|\widehat{f} - f\|_2^2 = \|(FB)^+(Fd + \eta)\|_2^2 + \|d\|_2^2$$

requires an orthonormal retained basis B, $d \perp \operatorname{ran} B$, and full column rank of FB. It exhibits why zero fitted residual cannot validate an infinite spectral conclusion. Such graph filters may guide witness search; exact discrete acceptance stays in $\mathcal{C}$.

TR250's cochain criterion $RP = I$, $I - PR = dH + Hd$ preserves cohomology. It does not preserve chromatic number: an edge and a point are both contractible but require two and one colors. It also does not automatically preserve the reduced free-homotopy words used in interface extraction. On a graph with two independent loops, a commutator has zero homology class but nontrivial reduced word. A coarse graph must therefore carry an explicit path/refutation transfer if it replaces that stage.

10.2 An upper-bound channel with a different logical role

A periodic ansatz takes $\Lambda = B\mathbb{Z}^2$, a bounded fundamental cell Q, and finitely many semialgebraic color cells with exact boundary ownership. If $D \geq \operatorname{diam} Q$ and $s > 0$ is a certified smallest singular value lower bound for B, any offending translated unit pair satisfies

$$|n| \leq (D + 1)/s.$$

Indeed $|Bn| \leq |x - y| + 1 \leq D + 1$. Only finitely many offsets remain; real quantifier elimination handles each equation $|x - y - Bn|^2 = 1$ with the actual strict/non-strict cell inequalities. Interval methods suffice with a positive clearance but need not terminate when one is approached and never attained. The interval $(0, 1)$ illustrates the boundary issue. No theorem here makes an arbitrary coloring periodic. Locally finite polygonal six-color ansatzes encounter the primary obstruction [11]; curved cells require a separate check of its broader map hypotheses.

11 Computational evidence and its limits

The authored checker uses only Python's exact `Fraction` arithmetic, finite enumeration, and SHA-256. Its field arithmetic is in $\mathbb{Q}(\sqrt{3}, \sqrt{11})$, with basis $1, \sqrt{3}, \sqrt{11}, \sqrt{33}$. It was actually run, with the following results:

Check	Result
Source build inventory	14 files match the pinned hashes
Spindle graph and placement	7 distinct vertices; 11 exact unit edges
All 3^7 spindle assignments	2187 tested; 0 proper
All 4^7 spindle assignments	16384 tested; 384 proper
Three-color badness	Minimum 1 monochromatic edge out of 11
Rational coordinate/polynomial boxes	All 7 rows pass, including intermediate factors
Uniform placement clearance	10^{-7} certified by the proved bound
Five-cycle six-position words	$5^6 = 15625$ tested; 0 survivors
Four-state rules on six positions	$3^6 = 729$ tested; 34 admissible words
Mixed-parity outsider words	Exactly 6; each isolated under the cross-word rule
Full spindle interpolation identity	All 2187 assignments checked

The first-violated-edge partition in Theorem 7.2, for the lexicographically ordered eleven spindle edges, has sizes

$$(729, 486, 324, 216, 216, 72, 72, 24, 24, 12, 12).$$

Their sum is 2187. Evaluation is an exact isomorphism of the quotient algebra, so checking the identity on every assignment is complete for this finite certificate. This is not a degree-truncated numerical moment test.

The independent local audit separately reports 4948 rational tests in its checker: 1025 equal-modulus identities, 1715 Hessian Gram identities, 100 mass-bound comparisons, and 2108 capped-residue arithmetic cases. Those are supplementary finite checks, not proofs by sampling of universally quantified lemmas. The general identities and inequalities require the analytic/algebraic proofs. We have not run a six-chromatic graph search, obtained a five-color UNSAT certificate, numerically realized arbitrary wild laws, or replayed Lean.

12 Assessment and a focused research program

Advantages. The source addresses unrestricted coloring through a specifically justified measurable bridge, and it extracts interfaces instead of assuming regular boundaries. Its final placement is exact and replayable. The Ram organization retains separate geometry, topology, measure, and logic obligations. The center and density bounds add explicit residual/mass constants; the circle tail bound states exactly which extra information would make selection effective. The finite algebra shows how full positivity can retain integral constraints that a fractional LP discards.

Costs and limitations. The proof has substantial nonconstructive dependencies: invariant means, compact towers, measurable kernels, limits, and compactness. A small exact terminal certificate does not certify these dependencies. The center constants can be ill-conditioned as $p \downarrow 1/2$ or R grows. Generic indicator functions offer no certified spectral tail rate. The complete finite algebra has exponential dimension. Basis changes, frame margins, machine precision, and cohomology preservation do not by themselves preserve color satisfiability. No performance advantage, historical novelty of the elementary estimates, or global chromatic closure is claimed.

Next mathematical work. The highest-value independent task is a pinned build and comparator replay, with the imported theorem chain, actual definition matching, and axiom output recorded. In parallel, an exact finite non-five-colorable witness would provide a different, compact verification route. For an effective analytic extraction, one would need quantitative transition rates, circle-tail bounds, separation margins, and finite selection bounds that hold under the original hypotheses. The scalar center module becomes computational only when its kernel masses and gradient errors are certified. Rational interval or Bernstein subdivision could enlarge the conservative placement radius without affecting any global claim. Weighted edge badness can be optimized on an actual witness while retaining an exact verifier. These are specific extensions, not promises that one finite refinement resolves the unrestricted problem.

13 Provenance and exact reading depth

Paths below are confidential local locators, not web-search queries. The private source authorship is Muralidhara Subbarao. Only the explicit passages listed for supplementary sources were used; there is no whole-corpus audit.

13.1 Pinned public-source snapshot

Root: `C:/Users/mural/OneDrive/Documents/OpenAI-722-math/`. Build directory relative to that root: `_sources/openai-math/preprints/The-Euclidean-plane-is-not-five-colorable-September-23-2026/build/`. Full reads: `main.tex`, `preamble.tex`, `references.bib`, and `sections/introduction.tex`, `sections/spectral.tex`, `sections/rigidity.tex`, `sections/transfer.tex`, `sections/palettes.tex`, `sections/transitions.tex`, `sections/interfaces.tex`, `sections/angular.tex`. Key stable labels include `thm:transfer`, `thm:rigidity`, `lem:spectral-location`, `lem:relative-singularity`, `lem:compact-tower`, `lem:multiple-averages`, `lem:palettes-continuous-factor`, `prop:palettes-forward`, `lem:palettes-density-proper`, `prop:palette-size`, `thm:cyclic-local-finiteness`, `prop:interface-cycle`, `lem:angular-certificate`, and `sec:angular-certificate-verification`. Figures were hash-checked, not relied on as independent geometric proof.

Catalogue: `00-Applicability-Catalogue/applicability.csv`, row `id=301`, `family=158`, `level=5`; all row fields read. Its direct match is ordinary coloring/transfer/certification, with an explicit boundary that the corpus supplies neither a non-five-colorable witness nor the Haar/palette theorems. Snapshot provenance: `_sources/snapshot.json` and `_sources/source-hashes.json`. Independent local review, read in full: `_audit/M301-parent-mathematical-review.txt`. The independent audit's finite-check files are `_audit/check_M301_parent.py` and `_audit/check_M301_parent.json`; the JSON result was read directly. Their execution was performed in the independent audit, not represented as a second independent run by this author.

Lean inspection, complete reads of the three local files: `_sources/lean-batch5/OAI/Geometry/PlaneColoring/Five.lean`, `_sources/lean-batch5/ComparatorChallenges/EuclideanFiveColor.lean`, and `_sources/lean-batch5/ComparatorChallenges/EuclideanFiveColor.json`. The catalogue mapping was supplied by the independent local audit from the pinned public `lean/formalization.yaml`. Imported dependencies were not exhaustively inspected or compiled.

13.2 Private primary report and discovered passages

1. **TR-item-094, full source including bibliography.** `C:/Users/mural/OneDrive/Documents/MyDocuments2026/res-rt/res-math/TR-item-094-astra.tex`. Exact section locators:

"Pinned Problem and Quantifiers", "Finite Witnesses", "An Exact Lower-Bound Certificate", "An Elementary Barrier to the Fractional Search", "Algebraic Completeness and Periodic Search Limits", and "Certified Ram Representation and Error Transfer". The date-qualified status, $8\sqrt{3}/\pi$ ceiling, algebraic completeness, safe symmetry reductions, and finite periodic offset bound were retained.

2. **TR248, selected definitions and proofs.** `C:/Users/mural/OneDrive/Documents/MyDocuments2026/res-rt/TechReps/FinalTechReps/TRs/TR248-Stable-Adaptive-Ram-Frames/TR248-Stable-Adaptive-Ram-Frames.tex`. Read lines 229–276 and 350–414, including `eq:leak`, `thm:finite`, `eq:modelerror`, `thm:perturb`, and physical-Gram normalization. No application of an unrestricted finite-dimensional reconstruction theorem is inferred.

3. **TR250, selected theorem/proof and examples.** `C:/Users/mural/OneDrive/Documents/MyDocuments2026/res-rt/TechReps/FinalTechReps/TRs/TR250-Compatible-Variational-Ram-Panels/TR250-Compatible-Variational-Ram-Panels.tex`. Read lines 737–812: `thm:homotopy`, `eq:chainmaps`, `eq:homotopy`, explicit interval refinement, and hole-filling counterexample. Only the cohomological statement is imported.

4. **TR190, graph filter definitions and limitations.** `C:/Users/mural/OneDrive/Documents/MyDocuments2026/res-rt/TechReps/FinalTechReps/TRs/TR190-GGRT/TR190-GGRT.tex`. Read lines 154–328: `def:atom`, `def:forward`, `def:inverse`, `prop:props`, and `sec:scope`. Its numerical verification table was read, not independently rerun. General unit-triangularity and scale-implied finite support were not adopted.

5. **Ram local moments, bounded chapter passage.** `C:/Users/mural/OneDrive/Documents/RamCalBooks/volumes/volume1-rev2/chapters/ch03-local-moments-rev2.tex`. Read lines 1–225, including `thm:ram-local`, `eq:ram-integral-remainder`, `eq:ram-remainder-bound`, `thm:degree-closure`, and the nonpositive-truncation example. Included chapter fragments were not fully read; only the displayed local identity/proof and cautions support Section 6.

6. **Application source, retained/discarded proof.** `C:/Users/mural/OneDrive/Documents/RamPatsCore50-TRs/TR-core-pat-5/TR-core-pat-5.tex`. Read lines 277–319, especially `prop:discarded` and `eq:discarded`. It supports the omitted-space warning, not any chromatic or Haar-rigidity theorem.

Discovery was through `_audit/corpus-core-map.txt` (E5, E6, E14), `_audit/corpus-foundations-map.txt` (F2, F4, F5 and surrounding scope warnings), and `_audit/corpus-applications-map.txt` (section E, C5). Those maps route to exact passages; they are not substitutes for them. TR165, TR187, TR188, TR191, TR205, and TR249 are mentioned by the private primary report but were not independently read here, and no theorem from them is claimed as an audited ingredient.

13.3 Reproducibility and public refresh

The author's checker is `M301-Plane-Coloring/evidence/check_m301_author.py`, with generated `M301-Plane-Coloring/evidence/checks.json`. It is invoked from the root with the bundled interpreter `C:/Users/mural/.cache/codex-runtimes/codex-primary-runtime/dependencies/python/python.exe`. Its finite checks are specified in Section 11; the rational proofs in the text are not replaced by code. The author evidence record is `_audit/M301-author-evidence.txt`.

Primary public metadata/abstracts were refreshed on October 7, 2026 for de Grey, Heule, Payne, Matolcsi et al., Sokolov–Voronov, Dúcz–Varga, and Dúcz. These refreshes were not full new proof

audits. Classical structural references below are retained from the fully read pinned bibliography and proof locators, not claimed reverified in full. Only public titles/IDs were queried. The source's discussion of other announced seven-color proofs is not used to establish any new status claim here.

14 Conclusion

M301 is an L5 match to the private finite-witness/certification material. The correct output of that applicability is a sharpened audit and several proved local quantitative modules, not an automatic global validation. The report preserves the arbitrary-coloring quantifier, the substantial Haar input, the exceptional-set discipline, and the discrete angular constraints. Its exact checks establish the finite terminal facts claimed for them. The global lower bound six remains a conclusion conditional here on the full pinned proof, with an existing formalization awaiting independent replay.

References

[1] OpenAI, *The Euclidean plane is not five-colorable*, September 23, 2026. Pinned commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. `https://github.com/openai/math/blob/adc7f1241b42e322a6451854ab7e4b4c146bf78a/preprints/The-Euclidean-plane-is-not-five-colorable-September-23-2026/build/main.tex`.

[2] OpenAI, *Formalization catalogue*, same pinned commit, entry for `OAI.Geometry.PlaneColoring.Five`. `https://github.com/openai/math/blob/adc7f1241b42e322a6451854ab7e4b4c146bf78a/lean/formalization.yaml`.

[3] N. G. de Bruijn and P. Erdős, *A colour problem for infinite graphs and a problem in the theory of relations*, Indagationes Mathematicae 13 (1951), 371–373, Theorem 1. `https://users.renyi.hu/~p_erdos/1951-01.pdf`.

[4] A. D. N. J. de Grey, *The chromatic number of the plane is at least 5*, 2018, v3. `https://arxiv.org/abs/1804.02385v3`.

[5] M. J. H. Heule, *Computing Small Unit-Distance Graphs with Chromatic Number 5*, 2018. `https://arxiv.org/abs/1805.12181`.

[6] K. J. Falconer, *The realization of distances in measurable subsets covering* $\mathbb{R}^n$, J. Combin. Theory A 31 (1981), 184–189. `https://doi.org/10.1016/0097-3165(81)90014-5`.

[7] M. S. Payne, *Unit distance graphs with ambiguous chromatic number*, Electron. J. Combin. 16 (2009), N31. `https://arxiv.org/abs/0707.1177v3`.

[8] M. Matolcsi, I. Z. Ruzsa, D. Varga, and P. Zsámboki, *The fractional chromatic number of the plane is at least 4*, arXiv:2311.10069, v4, March 27, 2025. `https://arxiv.org/abs/2311.10069v4`.

[9] Á. Dúcz and D. Varga, *A unit-distance graph in the plane with independence ratio below* 1/4, June 26, 2026. `https://arxiv.org/abs/2606.28157`.

[10] Á. Dúcz, *A note on geometric colorings of the Moser lattice*, June 10, 2026. `https://arxiv.org/abs/2606.12325`.

[11] G. Sokolov and V. Voronov, *On the chromatic number of the plane for map-type colorings*, arXiv:2502.01958 (2025), Theorems 1–2. `https://arxiv.org/abs/2502.01958`.

[12] L. Moser and W. Moser, *Solution to Problem 10*, Canadian Math. Bull. 4 (1961), 187–189. `https://doi.org/10.1017/S0008439500025765`.

[13] S. P. Townsend, *Colouring the plane with no monochrome units*, Geombinatorics 14 (2005), 184–193. https://www.jennysteve.net/MapColouring/2005paper.pdf.

[14] B. Bekka, P. de la Harpe, and A. Valette, *Kazhdan's Property (T)*, Cambridge University Press, 2008. Source locators D.2.2, D.3.1, G.2.1, G.2.2, G.5.1 refer to its cited February 23, 2007 author manuscript. https://doi.org/10.1017/CBO9780511542749.

[15] H. Furstenberg, *Ergodic behavior of diagonal measures and a theorem of Szemerédi on arithmetic progressions*, J. Analyse Math. 31 (1977), 204–256. https://doi.org/10.1007/BF02813304.

[16] R. J. Zimmer, *Extensions of ergodic group actions*, Illinois J. Math. 20 (1976), 373–409; *Ergodic actions with generalized discrete spectrum*, ibid., 555–588. https://doi.org/10.1215/ijm/1256049780; https://doi.org/10.1215/ijm/1256049648.

[17] A. Jamneshan, *An uncountable Furstenberg–Zimmer structure theory*, Ergodic Theory Dynam. Systems 43 (2023), 2404–2436; *Corrigendum*, 46 (2026), 211–217. https://doi.org/10.1017/etds.2022.43; https://doi.org/10.1017/etds.2025.10214.

[18] M. M. Crum, *On positive-definite functions*, Proc. London Math. Soc. (3) 6 (1956), 548–560. https://doi.org/10.1112/plms/s3-6.4.548.

[19] B. Deroin, V. Kleptsyn, and A. Navas, *On the question of ergodicity for minimal group actions on the circle*, 2008, Section 2.1. https://arxiv.org/abs/0806.1974v1.

[20] A. Hatcher, *Algebraic Topology*, Cambridge University Press, 2002, Theorem 1.9 and Section 1.A. https://pi.math.cornell.edu/~hatcher/AT/AT.pdf.

[21] T. Bourgeat, M. Heinrich, P. Melotti, and J.-M. Robert, *A probabilistic Hadwiger–Nelson problem*, 2015, https://arxiv.org/abs/1501.02441v1; H. Gwyn and J. Stavrianos, *A finite graph approach to the probabilistic Hadwiger–Nelson problem*, 2020 preprint, https://arxiv.org/abs/2008.07987v1.

[22] M. Subbarao, private technical reports TR-item-094, TR248, TR250, TR190, Ram local-moment chapter, and TR-core-pat-5. Exact paths, locators, dates, and reading boundaries appear in Section 13; these confidential sources are not public citations.